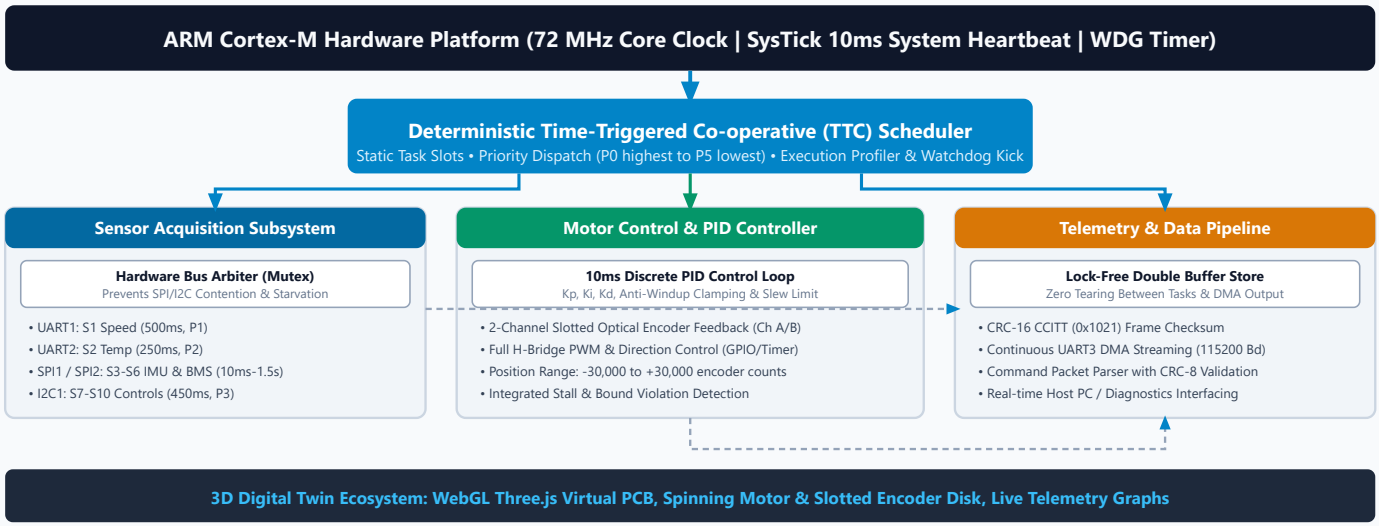
HoneyBadger ECU

Bare-Metal Time-Triggered Co-operative Automotive Firmware

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Task Code: FW9921
Firmware Name: HoneyBadger

1. BLOCK DIAGRAM OF PROPOSED CODE ARCHITECTURE



2. ARCHITECTURAL EXPLANATION IN SHORT & CRISP POINTERS

Deterministic Time-Triggered Execution

- **Zero RTOS Overhead:** Uses a pure co-operative static schedule table driven by a 10ms hardware SysTick timer, eliminating memory fragmentation and context-switch latency.
- **Priority Dispatching:** Strict priority bands (P0 to P5) guarantee time-critical tasks (IMU acquisition & PID loop) are dispatched before lower-frequency telemetry.
- **Watchdog Hardening:** SysTick kicks the hardware watchdog (IWDG) only after verifying that all scheduler task execution windows completed within their allotted limits.

Multi-Sensor Bus Arbitration

- **Shared-Bus Protection:** A dedicated Bus Arbiter guards SPI1, SPI2, and I2C1 using mutex state locks, preventing conflicting transactions between sensors S3–S10.
- **Starvation-Free Non-Blocking:** Sensor tasks that find a bus locked gracefully yield and re-queue without halting the CPU or starving other peripherals.
- **Configurable Polling Rates:** Asynchronous sensor sampling spans 10ms (IMU accelerometer) up to 1500ms (battery pack current monitor).

High-Precision PID Motor Control

- **10ms Control Loop:** Discrete-time PID algorithm calculates proportional, integral, and derivative corrections for precise angular position targeting.
- **Dual Optical Encoder Feedback:** Simulates and monitors Channel A/B 90° phase-shifted quadrature pulses with edge-triggered position decoding.
- **Anti-Windup Clamping:** Eliminates integral saturation during high slew rates and automatically applies electronic braking upon reaching setpoints.

Data Pipeline & Integrity Validation

- **Lock-Free Double Buffering:** A ping-pong double buffer ensures sensor acquisition tasks can write snapshots concurrently while UART3 DMA streams data tear-free.
- **Automotive Frame Safety:** All outbound packets are sealed with CRC-16-CCITT (0x1021) checksums; inbound motor commands require CRC-8 validation.
- **3D Digital Twin Co-Simulation:** WebGL-based digital twin models the circuit board, sensors, motor, and encoder disk in real-time at 60 FPS.

What makes this solution special: Its browser-based 3D digital twin mirrors scheduler timing, sensor-bus activity, PID motor response, and CRC-checked telemetry in real time, making firmware behavior observable without physical ECU hardware.